



PowerShell Conference Europe

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3

2

1



PowerShell Conference Europe

Azure Machine Management at Scale: Govern, Inventory, and Configure Your Servers

Steven Bucher

Many thanks to our sponsors:

Devolutions



Steven Bucher

- Previously PowerShell & SSH Team

- AI Shell
- PSReadLine
- SSH

- Worked on Machine Management in Azure for last year

- Also happy to talk about Azure Arc

- Currently working on Azure Governance and Resource Manager Team

- 5th year attending PSConfEU!



SPEAKER



<https://www.linkedin.com/in/stevenabucher/>

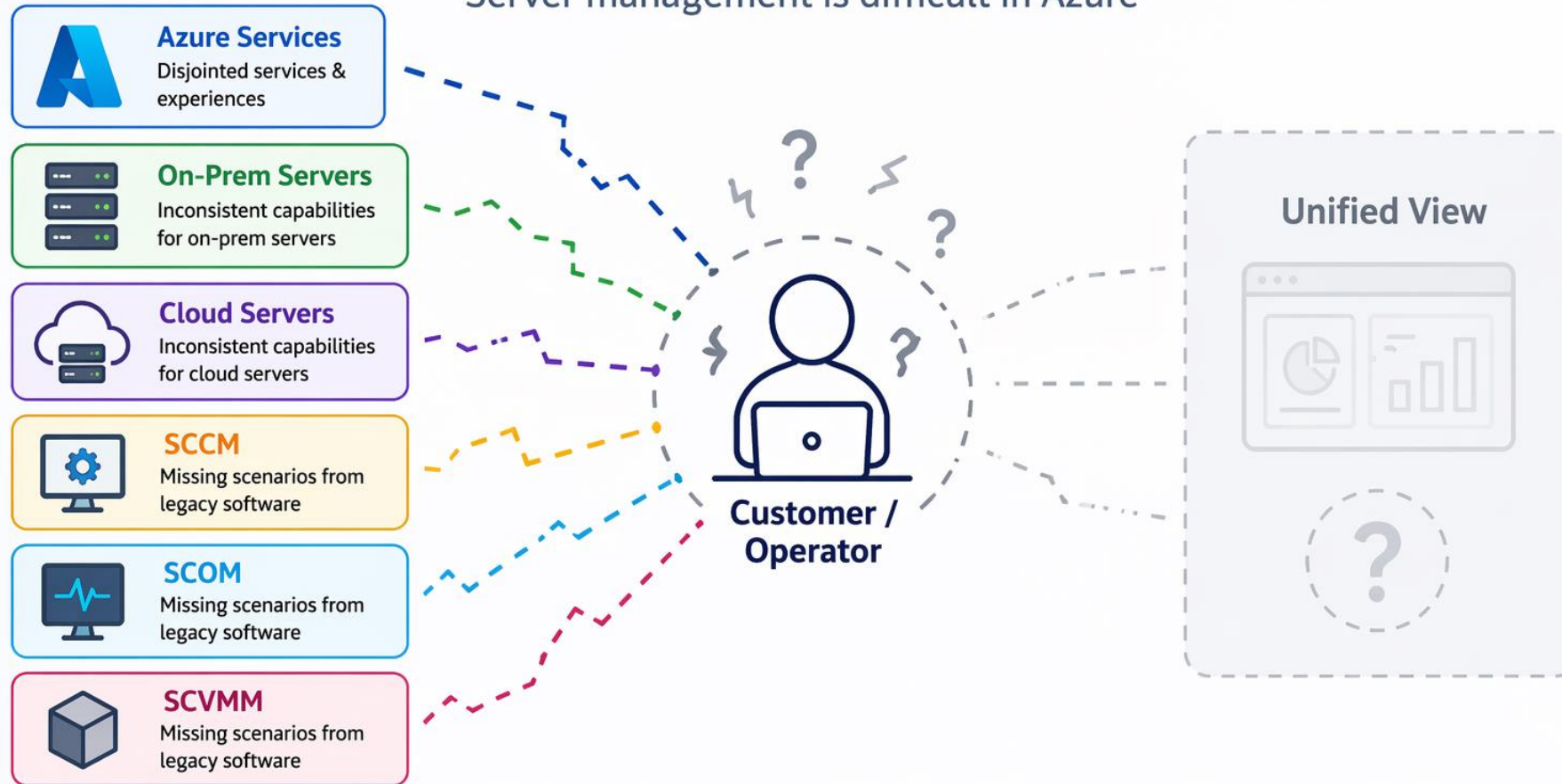
Current problem

Server management is difficult in Azure

- Disjointed services & experiences
- Inconsistent capabilities for cloud and on-prem servers
- Missing scenarios from legacy software like SCCM, SCOM, SCVMM

CURRENT PROBLEM

Server management is difficult in Azure



Fragmented, siloed services → **No single pane of glass**

Customers today navigate a fragmented, siloed set of services with no unified view.

Essentials Machine Management (EMM) Bundle

- Enables these services and configures base defaults properly across Azure VM & Arc-enabled VMs

Current experience

- Disjointed experience when enabling services individually
- Time consuming for customers and not easy
- Customer needs to know different/key Azure services to add
- Pricing complexity and lack of transparency

With EMM

- Easier to enable all necessary management services
- Transparent pricing and cheaper for Arc customers

Public Preview: Essential Machine Management

Centralized onboarding

Subscription-level onboarding experience that simplifies **machine** management at-scale

Resource Types

Azure VMs and **Arc-enabled servers** supported

Services Included

Auto-enrolls machines into **Monitor, Update Manager, Machine Configuration, and Change Tracking & Inventory**

Pricing

Free for Azure VMs & Arc Servers with WS Software Assurance/ESU/PayGo

\$9 / Arc Server / month*

***Currently free at no additional cost; billing will begin prior to GA**

Pricing

During the initial phase of public preview, EMM is free for all machines. Starting ~July 1st, billing will begin for Arc Servers.

	Currently Available	EMM Bundle
Management Services	A-la-carte	EMM Offer (\$9 per Arc Server, free for Azure VM)
Change Tracking and Inventory (**Ingestion rates are outside of EMM offer)	\$6/Server/Month	✓
Azure Policy Machine Configuration	Free with Azure VMs	✓
Azure Update Manager	\$5/Server/Month Free with Azure VMs	✓
Azure Monitor	\$2.30/GB/day	<i>Metrics-based VM Insights & Recommended alerts</i>

EMM is included at no additional cost for Windows Server Software Assurance, Windows Server PayGo, Windows Server Extended Security Updates, and Azure Local machines

****Log ingestion rates for Essential Machine Management are charged separately****

Demos

EMM!



Management Services

Feature	Configuration details
Azure Monitor	- Installs Azure Monitor agent - Collects standard set of performance counters (metrics). - Configures recommended alerts
Azure update manager	- Installs extension (within next 24 hours) (Microsoft.CPlat.Core.LinuxPatchExtension or Microsoft.CPlat.Core.WindowsPatchExtension) - Periodic assessment enabled.
Change tracking and inventory	- Install extension (Microsoft.Azure.ChangeTrackingAndInventory-Windows or Microsoft.Azure.ChangeTrackingAndInventory.ChangeTracking-Linux) - Uses Log Analytics workspace specified in onboarding. - Collects basic files and registry keys.
Machine Configuration	- Installs the Machine Configuration extension (Microsoft.GuestConfiguration.ConfigurationforWindows or Microsoft.GuestConfiguration.ConfigurationforLinux) - Applies the Azure Security Baselines in <i>Audit-only</i> mode

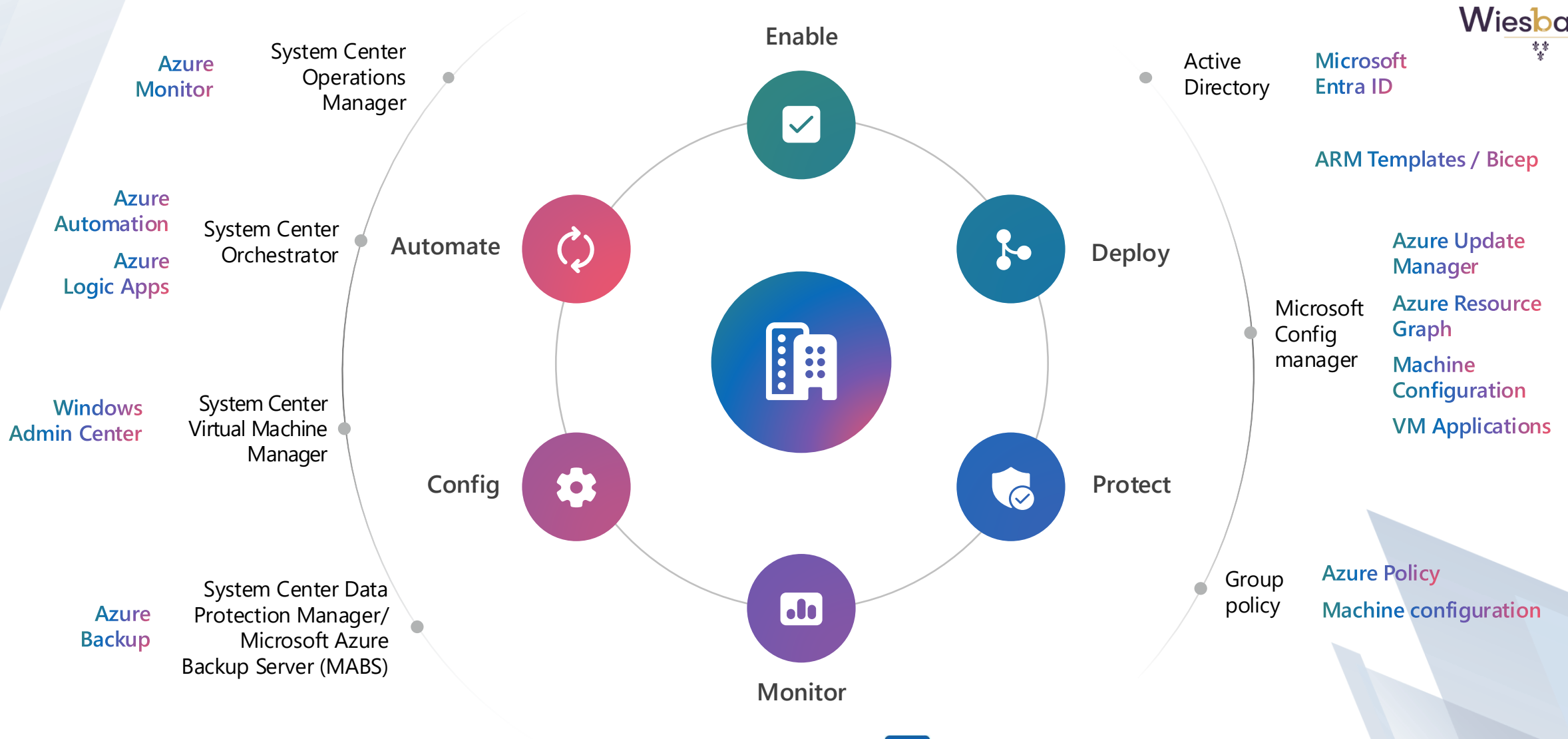
Pre-reqs

- Resources
 - [Log Analytics workspace](#) to collect log data collected from the machines
 - [Azure Monitor workspace](#) to collect metrics data collected from the machines
 - [User assigned managed identity](#) with **Contributor permission** for the subscription
- Permissions on the subscription
 - Essential Machine Management Administrator role
 - Managed Identity Operator roles
 - Resource Policy Contributor role

Azure's strategy for managing hybrid environments



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Machine Configuration

What is Machine Configuration?

- Azure service that provides capability to *audit* or *configure* operating system settings as code.
 - both for machines running in Azure and hybrid Arc-enabled machines
- Utilize Azure Policy to provide at scale configuration across your Azure environment.
- Yes you can do this all through PowerShell 😊
 - Lots on content online on how to do custom configurations!

Machine Configuration Improvements

- UX improvements
 - Prerequisite installation
 - Improvements to report metadata
- Baseline customization is GA!
 - CIS for baseline for Windows is now in public preview
 - Platform UX improvements

Baseline

Standard	Status	Notes
CIS Benchmarks for Linux	Generally Available	Expanded distribution coverage since Public Preview. See the full list of supported distros in the official documentation .
[NEW!] CIS Benchmarks for Windows	Public Preview	Initial release covers L1 settings for WS2025 Domain Controller and Member Server roles.
Azure Compute Security Baseline for Windows	Generally Available	Now supports customization for Windows Server 2016 and 2019, in addition to 2022 and 2025.
Azure Compute Security Baseline for Linux	Generally Available	Aligned with Azure Compute recommendations across supported Linux distributions.

Demos



Learn more!

- Customizable Baseline GA: <https://aka.ms/BaselineCustomizationGA>
- CIS for Windows: <http://aka.ms/cis4windows>



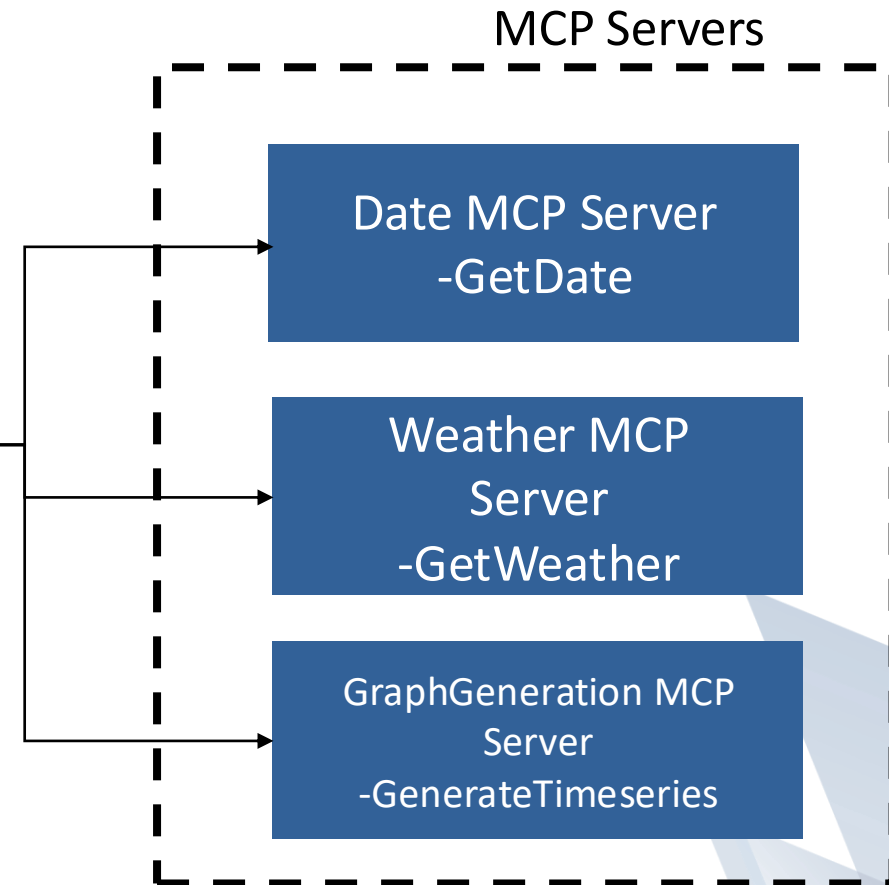
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Azure Resource Manager MCP Server

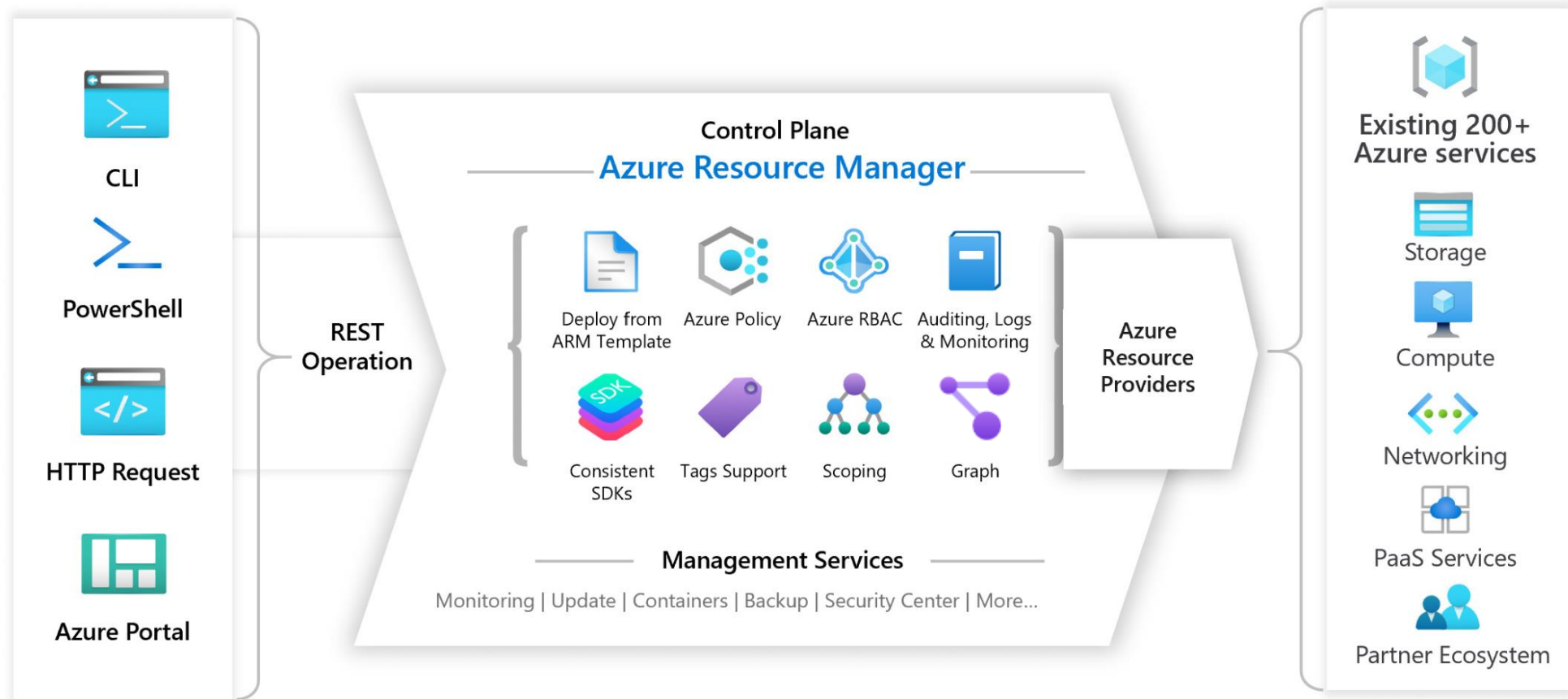
Model Context Protocol

Standard protocol for AI to use repeatable tools

“What is the weather today and for the next week”



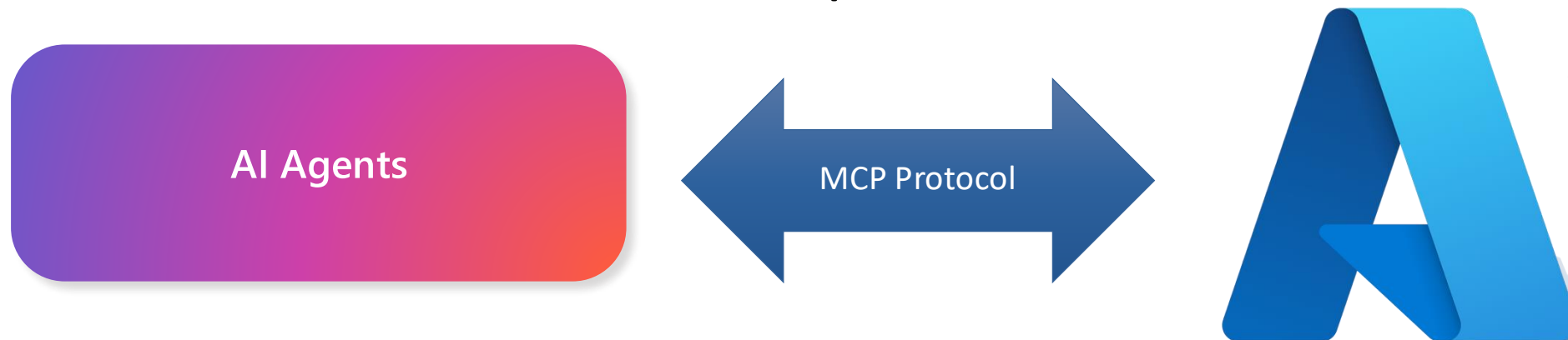
What is Azure Resource Manager?



Azure Resource Manager MCP Server

- A centralized hosted MCP Server to help you get reliable data about your Azure estate
- Connects your agents to real time data of **any** resource type in Azure
- Able to get compliance data back about your environment

Check it out at <https://aka.ms/ARMMCP>

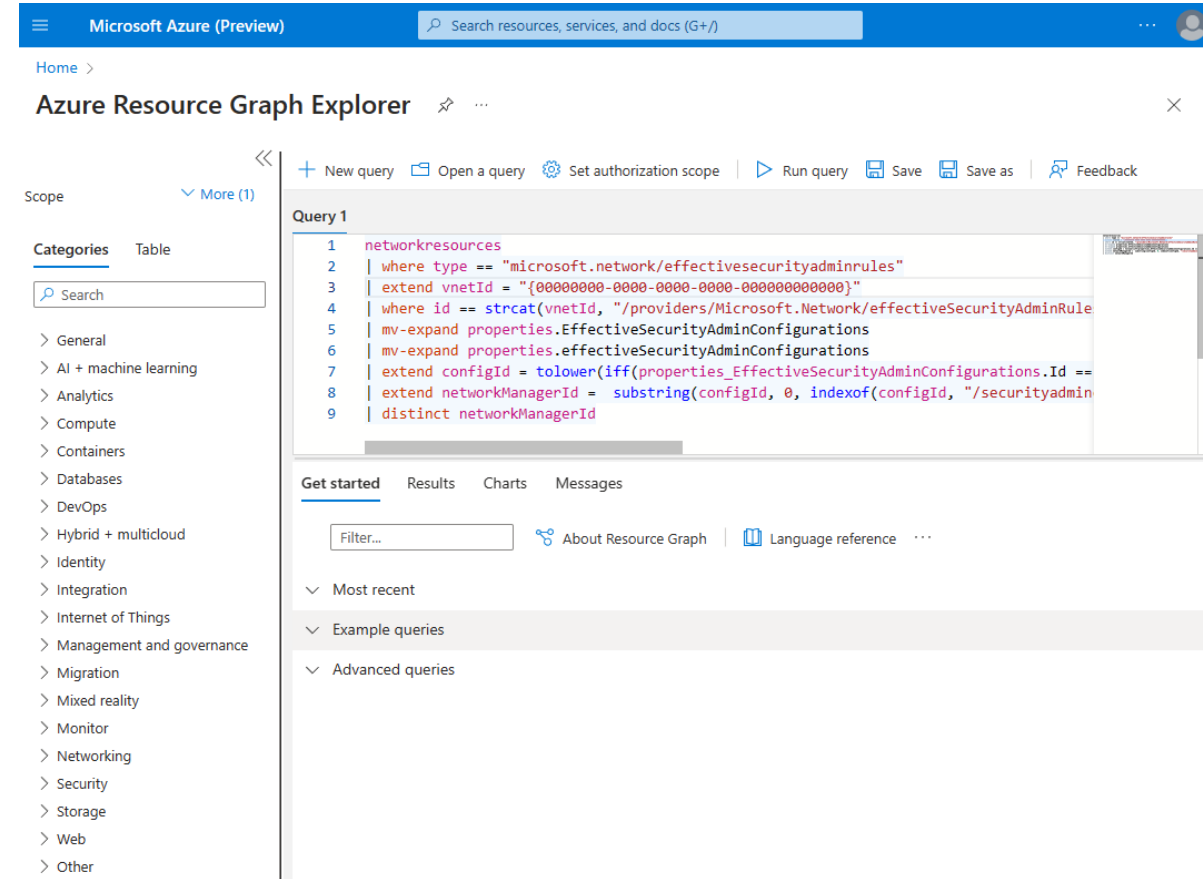


Read Tools

Utilizes Azure Resource Graph queries

- Generate_query
- Validate_query
- Execute_query

Most of this you don't have to worry about 😊



Demos

Azure
Management
MCP READ



Deployment Tool list

Foundational tools for Write scenarios

- `create_or_update_resource_group`
- `create_deployment`
- `get_deployment_status`
- `cancel_deployment`
- `whatif_deployment`
- `get_whatif_deployment_status`

Intent is processed and created by the user's local client LLM



User's Local Client (LLM)



1. Intent

User defines the desired outcome in natural language.
(e.g., *create another storage account with same configuration*).



2. ARM Template Generation

The local LLM generates an ARM template (JSON) based on the intent.



3. Validation / Whatif

Validate the template and run What-If to preview changes.



No errors
(Looks good)



Errors found
(Fix and regenerate)



4. Deploy

Deploy the ARM template to Azure.



You can cancel deployment at any time.



5. Check Deployment Status

Check the deployment status until completion.



Succeeded



In progress



Failed

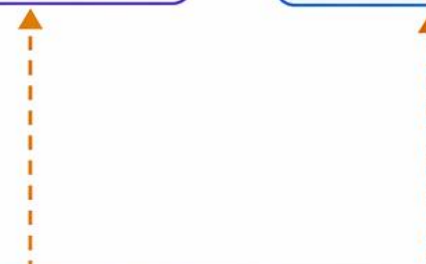


Canceled



Errors found?

Regenerate ARM template and re-validate until no errors.



Cancel at any time: You can cancel the deployment at any time during the process (validation, deployment, or while checking status).

→ Normal flow

- - -> Loop on errors

- - -> Cancellation option (available at any time)

Demos

Azure
Management
MCP Stage
scenarios



How to use it!

- Repository: <https://aka.ms/ARMMCP>
- Install in VSCode: <https://aka.ms/JoinARMMCP>
- Tech community blog:
<https://techcommunity.microsoft.com/category/azure/blog/azuregovernanceandmanagementblog>

Part of the Azure MCP Server!

- <https://github.com/microsoft/mcp>

Q&A

15 minutes



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GIVE ME FEEDBACK



Azure Machine Management at Scale: Govern, Inventory, and
Configure Your Servers

Connect with me!



<https://www.linkedin.com/in/stevenabucher/>